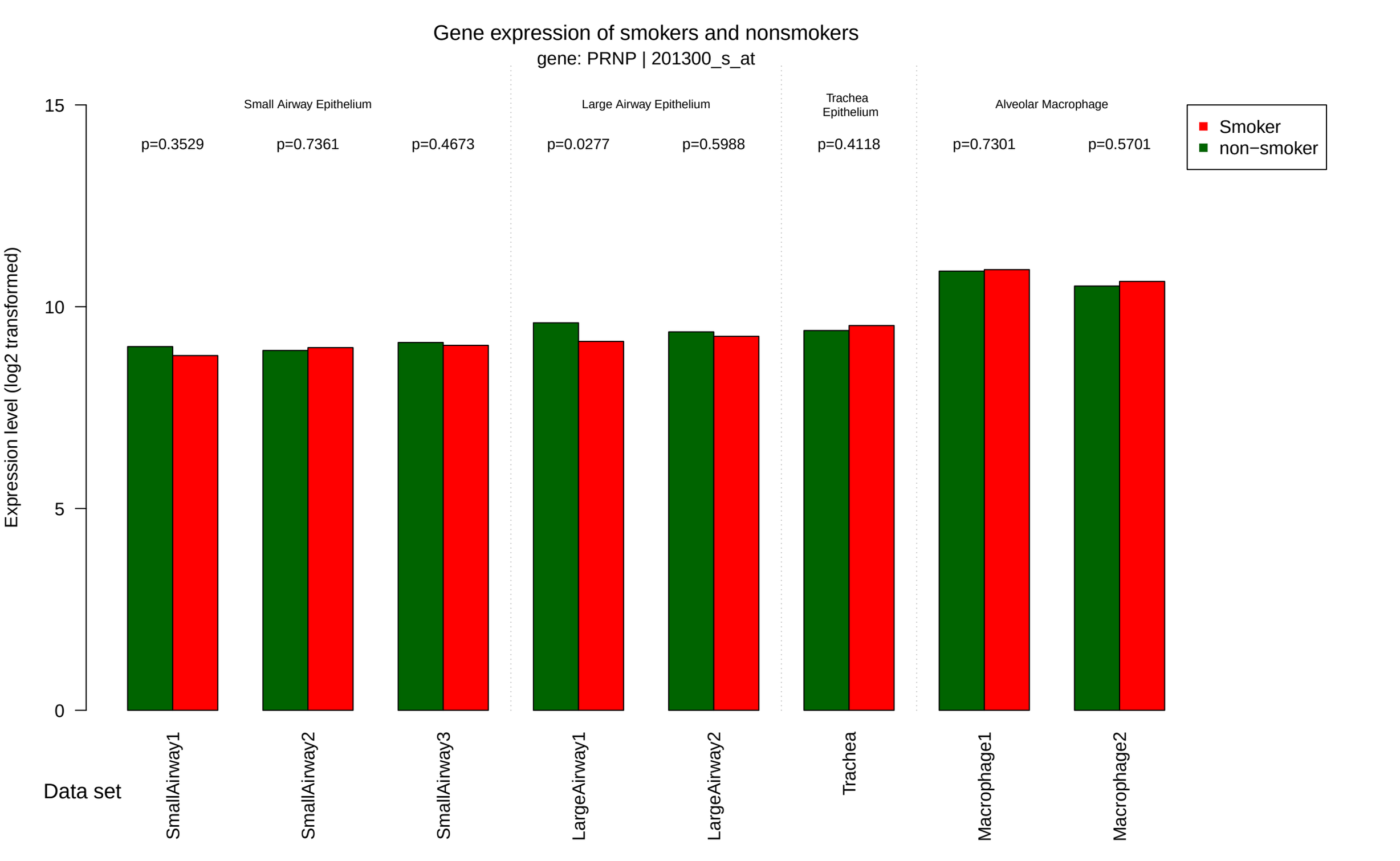
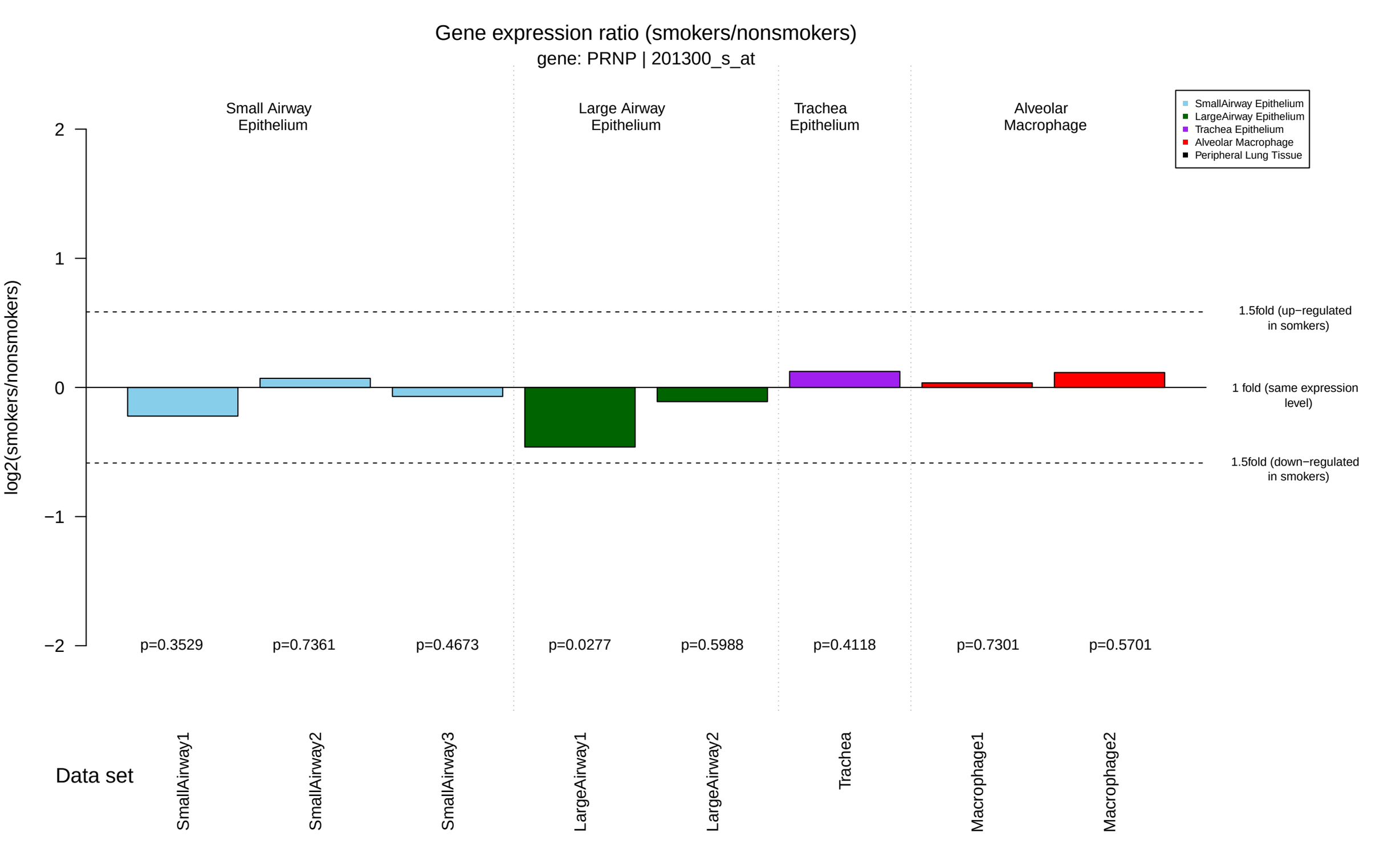
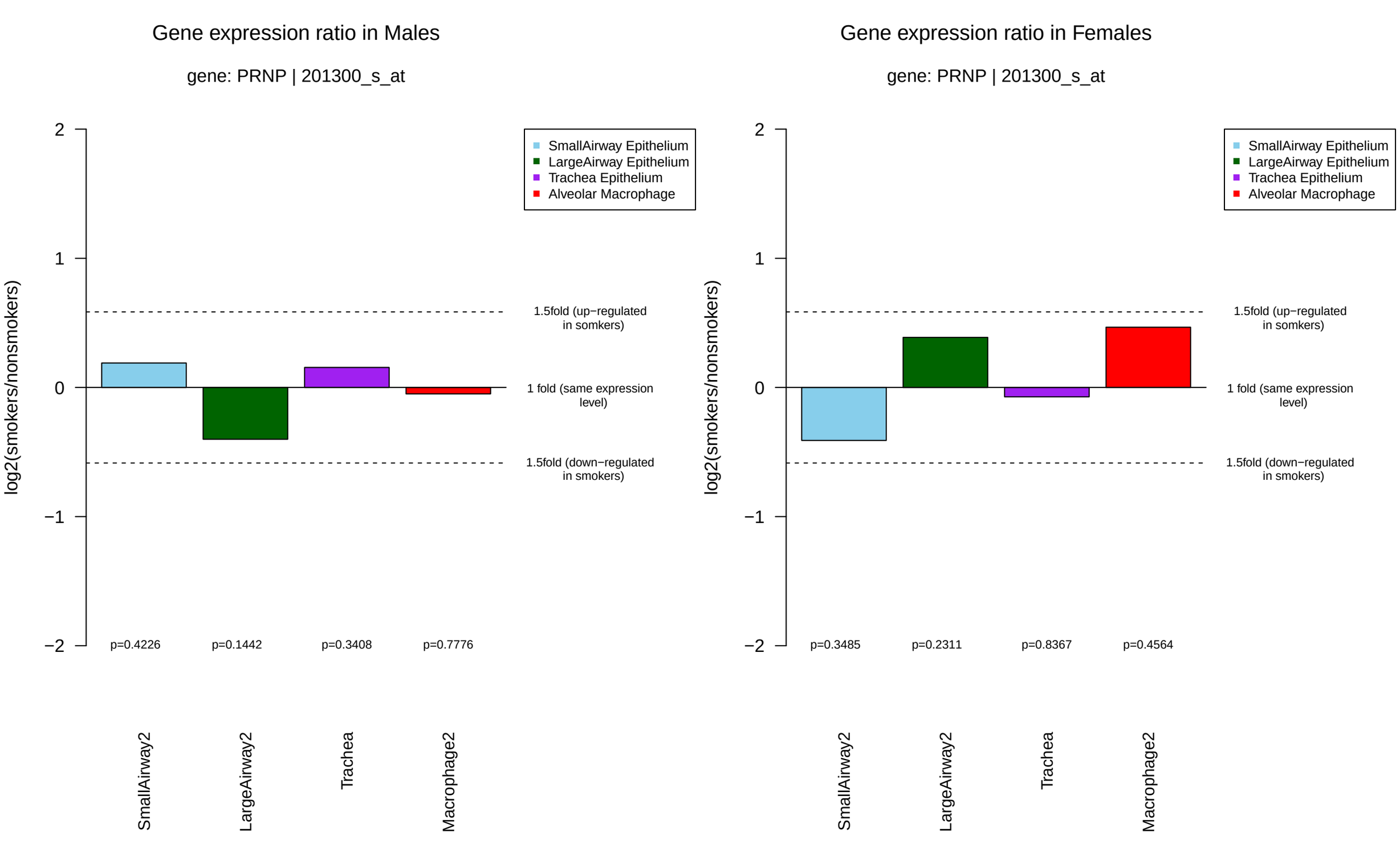
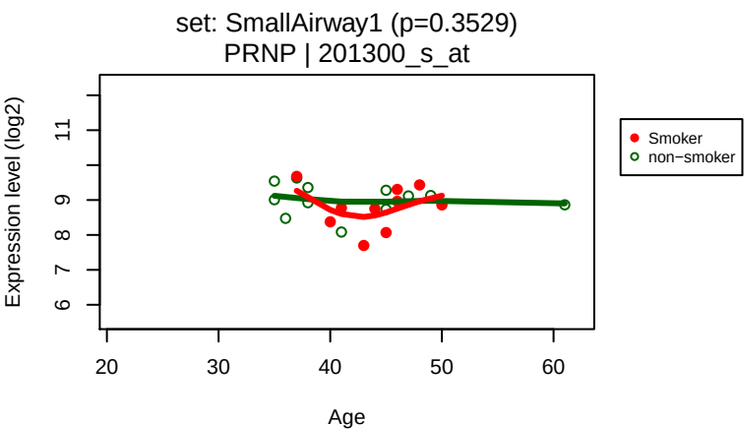
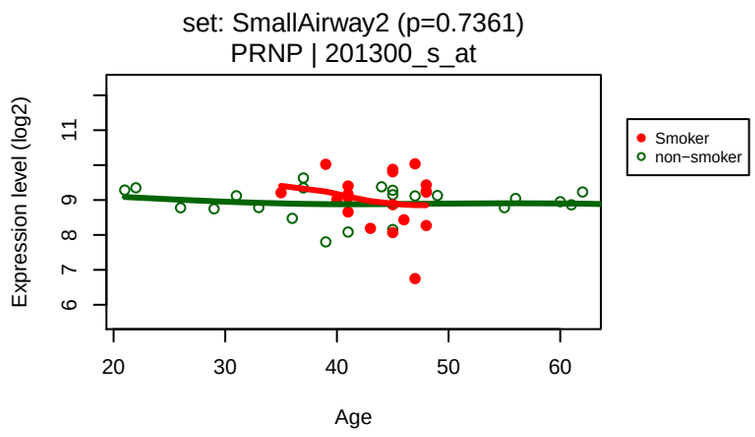
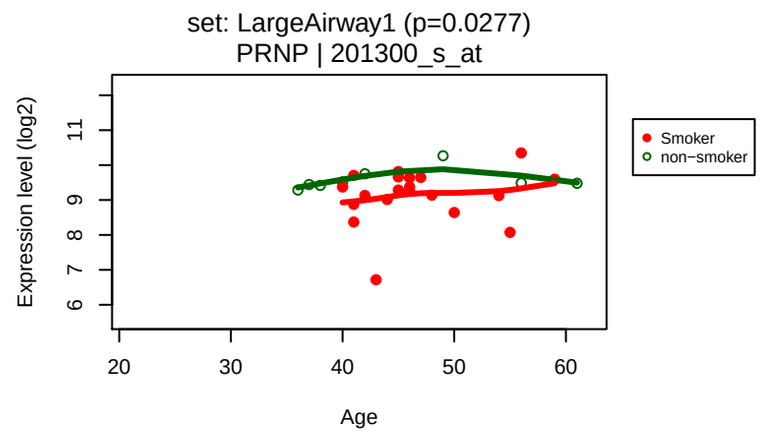

Gene expression of smokers and nonsmokers
gene: PRNP | 201300_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3529
p=0.7361
p=0.4673
p=0.0277
p=0.5988
p=0.4118
p=0.7301
p=0.5701
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

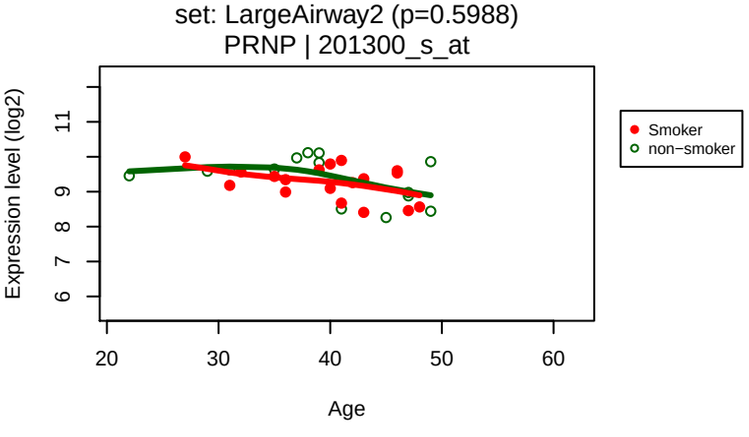

Gene expression ratio (smokers/nonsmokers)
gene: PRNP | 201300_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3529
p=0.7361
p=0.4673
p=0.0277
p=0.5988
p=0.4118
p=0.7301
p=0.5701
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

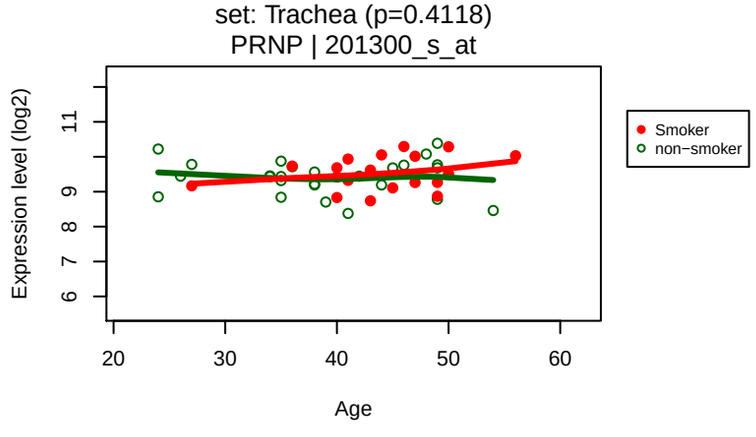

Gene expression ratio in Males
gene: PRNP | 201300_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4226
p=0.1442
p=0.3408
p=0.7776
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRNP | 201300_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3485
p=0.2311
p=0.8367
p=0.4564
SmallAirway2
LargeAirway2
Trachea
Macrophage2

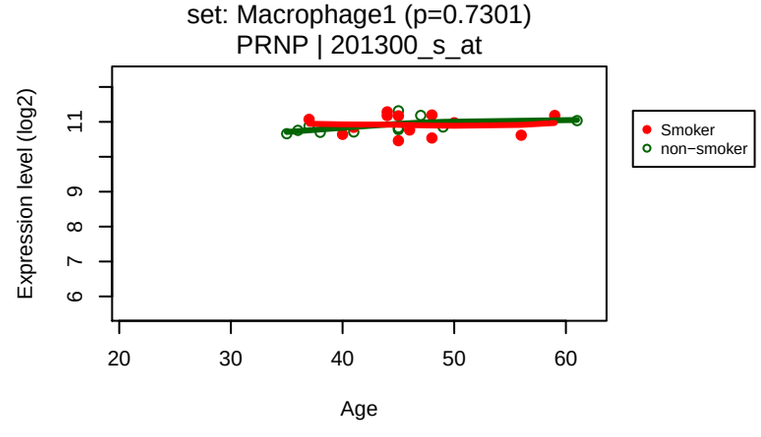

set: SmallAirway1 (p=0.3529)
PRNP | 201300_s_at
Expression level (log2)
Age
Smoker
non−smoker

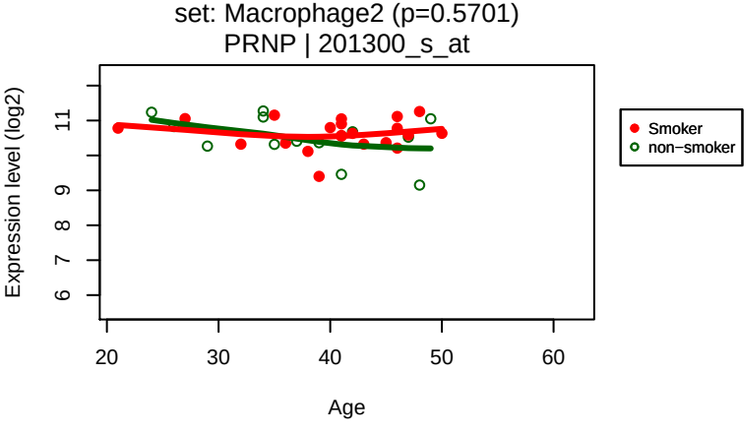

set: SmallAirway2 (p=0.7361)
PRNP | 201300_s_at
Expression level (log2)
Age
Smoker
non−smoker

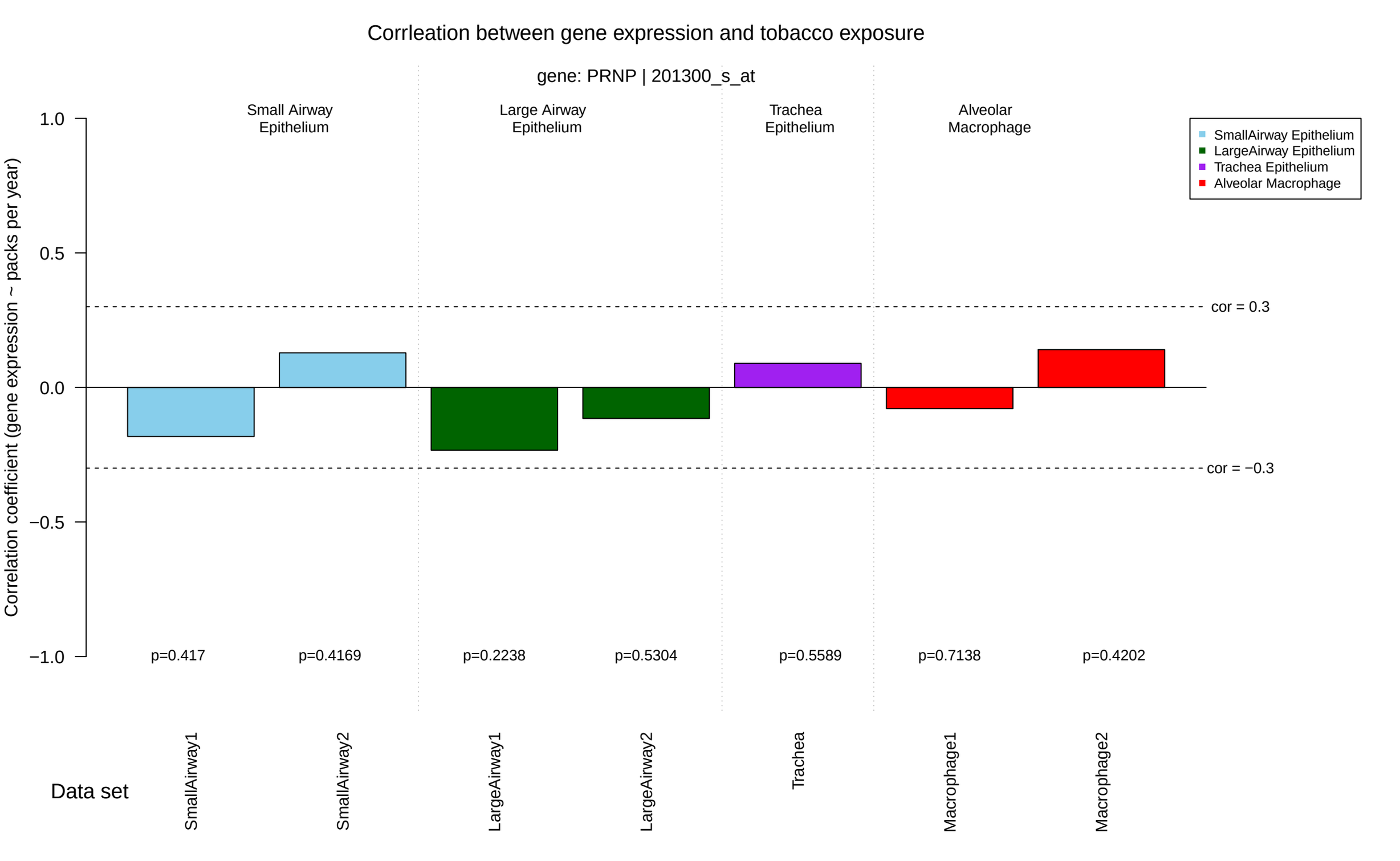

set: LargeAirway1 (p=0.0277)
PRNP | 201300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5988)
PRNP | 201300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4118)
PRNP | 201300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7301)
PRNP | 201300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5701)
PRNP | 201300_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRNP | 201300_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.417
p=0.4169
p=0.2238
p=0.5304
p=0.5589
p=0.7138
p=0.4202
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2